

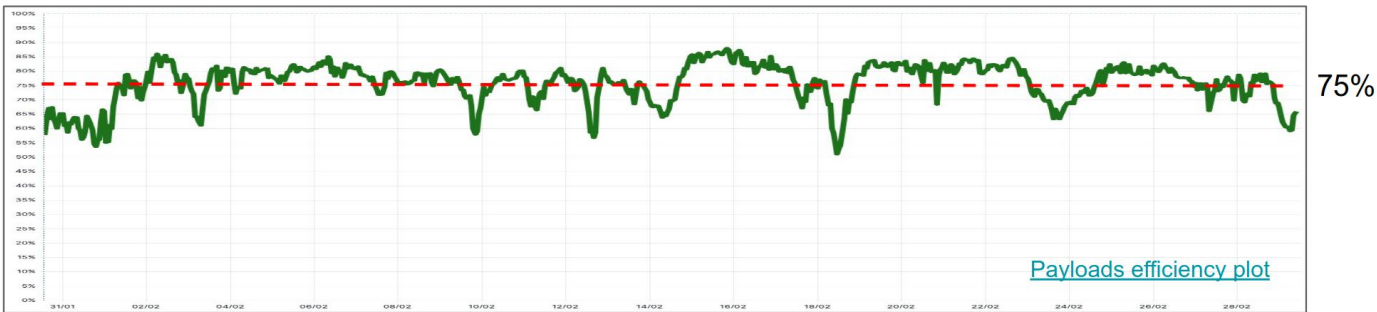
CMS Pilot Overloading

- Submission Infrastructure way of helping with CMS efficiency ([Details](#))
 - Using sites efficiently is key for the WLCG Computing Resources Scrutiny Group

Scheduling efficiency (used CPU cores / total allocated CPU)



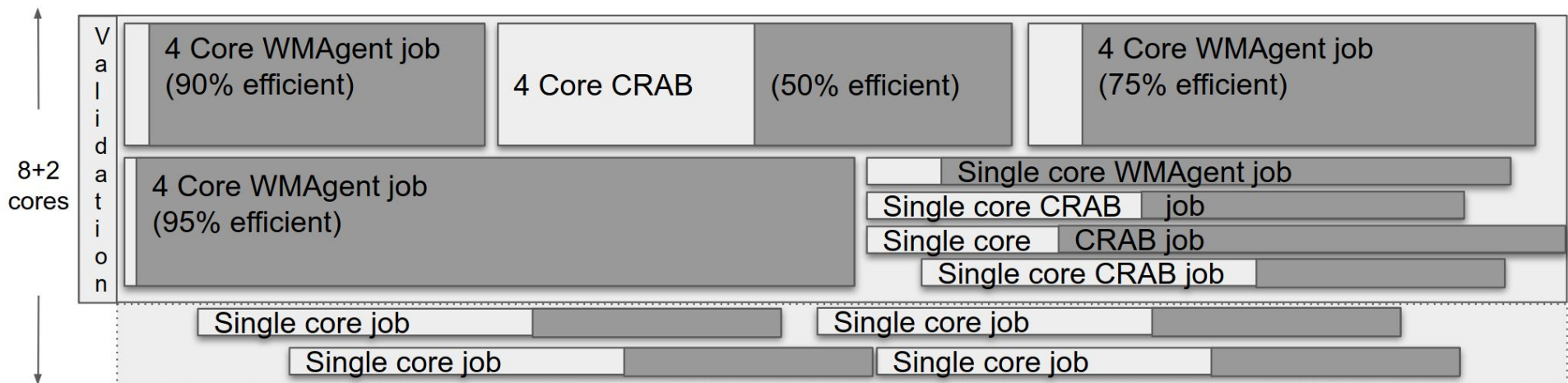
Payload efficiency from the standard job dashboard for [production jobs](#) at T1s



Strategy to recover unused CPU cycles

Idea: Improve CPU utilization efficiency by pushing more work into the same pilot envelope

- A simple approach is to “enlarge” our pilots so that they accept more payload jobs into the same resources acquired from the site
 - Trivial to implement and test from CMS Submission Infrastructure side
- Principle: we want to recover the unused CPU, not gain opportunistic cycles!
 - Moderate overloading: add 25% extra CPU cores and memory to the nominal values of our standard 8-core pilot.
 - Provides 2 extra cores, e.g. available to run additional CRAB or production payloads



Monitoring this relies on EGI **monthly** accounting

Pilot efficiency calculator

- I worked on a tool to calculate the efficiency of pilots:

```
[mmascher@vocms0209 tools]$ ./efficiency.py -h
usage: efficiency [-h] [--verbose] [--aggregate] [--cpus CPUS]
                [--gridtype GRIDTYPE]
                ACTIVITY_FILES [ACTIVITY_FILES ...]
```

Iterates over the list of condor_activity files to perform efficiency calculations and print the results.

positional arguments:

ACTIVITY_FILES	List of condor activity files to iterate over
----------------	---

optional arguments:

-h, --help	show this help message and exit
--verbose	Verbose mode
--aggregate	Aggregate the information of all the pilots and print a summary
--cpus CPUS	Number of CPUs of the pilot, or "auto" for whole node pilots
--gridtype GRIDTYPE	Type of CE. Can either be "condor" or "arc"

Pilot efficiency calculator

- Allows to check pilot efficiency daily for overloaded resources live by parsing condor event logs:

```
...
001 (9828657.001.000) 2024-02-14 13:58:24 Job executing on host: condor cmsosgce3.fnal.gov
cmsosgce3.fnal.gov:9619
...
005 (9828657.001.000) 2024-02-21 12:27:41 Job terminated.
(1) Normal termination (return value 0)
    Usr 293 00:25:10, Sys 4 01:00:18 - Run Remote Usage
    Usr 0 00:00:00, Sys 0 00:00:00 - Run Local Usage
...
```

- Have to parse job stdout and stderr to get pilot CPU cores for whole node
 - Feature HTCSS release 23.6 will put the number of cores used by the pilot in the event log

Pilot efficiency calculator

- Can either print overall efficiency:

```
[mmascher@vocms0209 tools]$ export  
LOG_DIR="/data/glideinmonitor/upload/CERN/user_fecmsglobal/glidein_gfactory_instance/entry_CMSHTPC_T1_US_F  
NAL_condce2_whole/"  
[mmascher@vocms0209 tools]$ ./efficiency.py  
/data/glideinmonitor/rsync/*/user_fecmsglobal/glidein_gfactory_instance/entry_CMSHTPC_T1_US_FNAL_condce2_w  
hole/* --cpus auto --aggregate  
Total efficiency 93.03%
```

- Or single pilot efficiency:

```
[mmascher@vocms0209 tools]$ ./efficiency.py  
/data/glideinmonitor/rsync/*/user_fecmsglobal/glidein_gfactory_instance/entry_CMSHTPC_T1_US_FNAL_condce2_w  
hole/* --cpus auto | tail -n 5  
Overloaded Job 6502336.000.000 lasted 597420.0. Efficiency is 93.88%  
Overloaded Job 6503002.000.000 lasted 587997.0. Efficiency is 95.97%  
Overloaded Job 6504867.005.000 lasted 599375.0. Efficiency is 95.86%  
Overloaded Job 6506095.000.000 lasted 600260.0. Efficiency is 93.84%  
Overloaded Job 6506192.000.000 lasted 601389.0. Efficiency is 96.73%
```

- Plan to use it to populate CMS MonIt Elastic Search database